Internship Report

1. Introduction

During my cybersecurity internship, I worked on a foundational project focused on understanding and implementing basic network security measures. The purpose of this task was to build practical knowledge of how small networks can be protected against common cyber threats and how security tools are used in real-world environments.

The main objective of this project was to study different types of network threats such as viruses, worms, trojans, and phishing attacks, and understand core security concepts including firewalls, encryption, and secure configurations. In addition, I set up a simple network environment, configured basic protection mechanisms such as enabling the firewall and securing Wi-Fi encryption, and used Wireshark to monitor and analyze network traffic.

This task aimed to strengthen my understanding of how network attacks occur, how suspicious traffic can be identified, and how fundamental security controls help protect systems and data from unauthorized access.

2. Task Overview

2.1 Task 1: Introduction to Network Security Basics

Detailed Description – Task 1: Introduction to Network Security Basics

Objective of the Task

The objective of this task was to understand the basic concepts of network security and learn how to secure a small network from common cyber threats. I aimed to gain practical knowledge about different types of network attacks and implement simple security measures to protect systems from unauthorized access. I also wanted to learn how to monitor and analyze network traffic to identify normal and suspicious activities.

Tools and Methods Used

Tools Used:

- Windows Defender Firewall
- Wireshark

- Home Wi-Fi router

Methods Used:

- Researching common network threats such as viruses, worms, trojans, and phishing attacks.
- Configuring firewall and router security settings.
- Capturing and analyzing live network traffic using Wireshark.
- Applying filters to observe specific protocols like HTTP, DNS, and TCP.

Steps Taken to Complete the Task

Task 1 – Introduction to Network Security Basics

Setting Up the Network Environment

To complete this task, a simple virtual network environment was created using VirtualBox. The environment consisted of two virtual machines: an Ubuntu system configured as a router and a Kali Linux system configured as a client device. This setup simulated a small home network where one device acts as a router while another device connects to it as a client.

The Ubuntu router was configured with two network interfaces. The first interface provided internet access through NAT, while the second interface was connected to an internal network that allowed communication with the client device. This configuration enabled the router to forward traffic between the internal network and the internet.

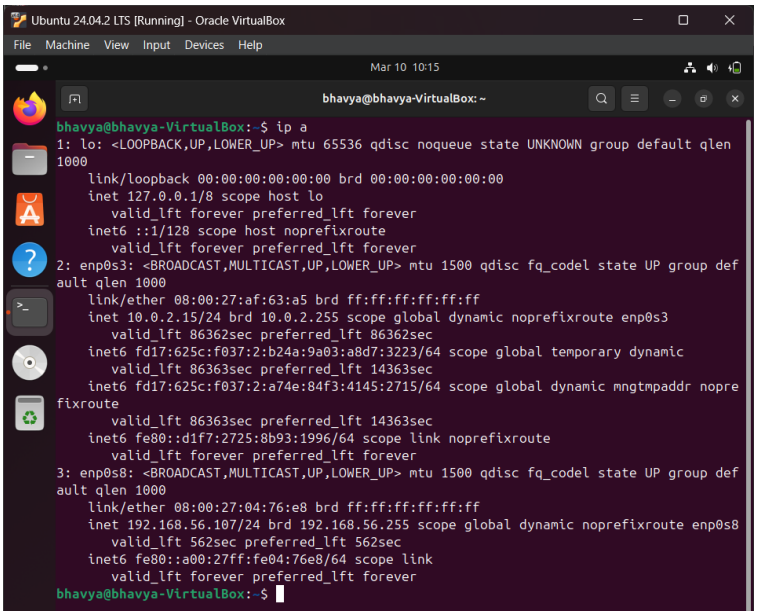


Figure 1: Network interface configuration on the Ubuntu router showing internet and internal network interfaces.

Enabling IP Forwarding

IP forwarding was enabled on the Ubuntu router so that it could route packets between the internal network and the external network. This allowed the router to forward traffic from the client device to the internet and back.

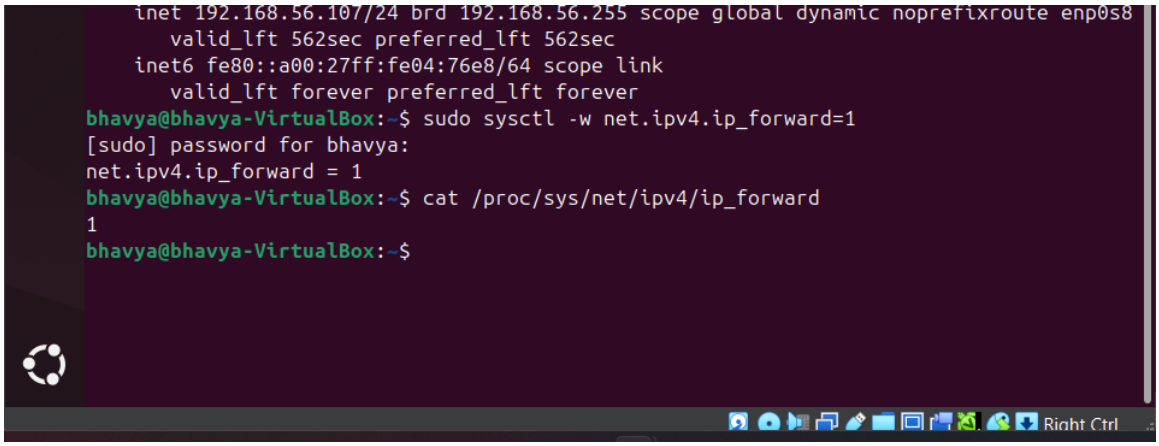


Figure 2: IP forwarding enabled on the router to allow packet routing between networks.

Client Network Configuration

The Kali Linux virtual machine was configured as the client device in the network. The client was connected to the internal network and assigned an IP address within the same subnet as the router. This allowed the client device to communicate with the router.

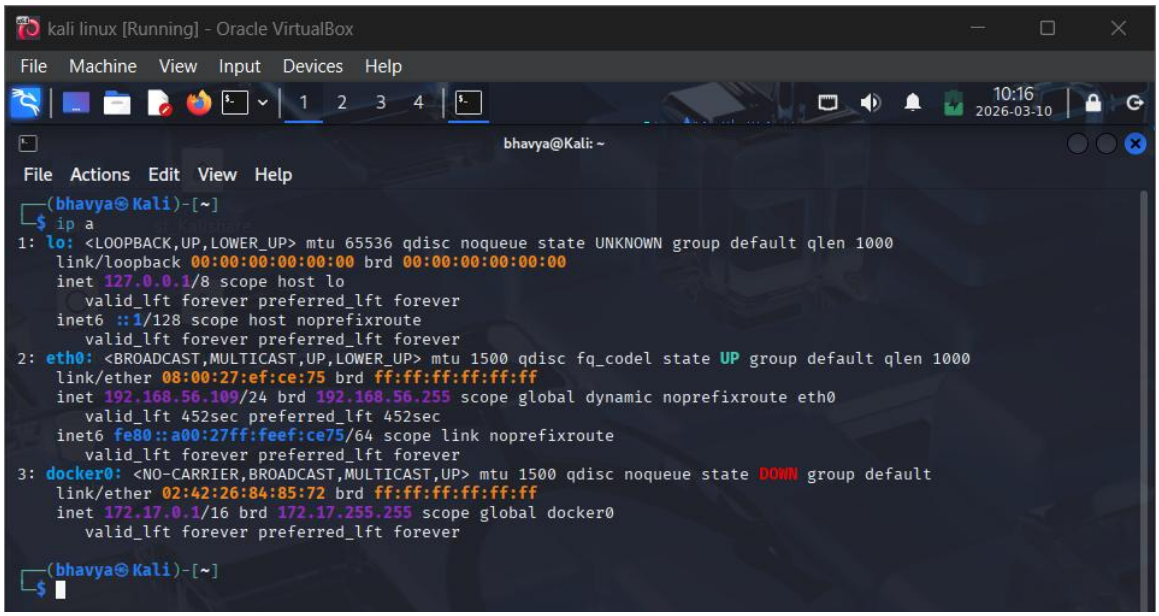


Figure 3: Network interface configuration on the Kali Linux client.

Configuring the Default Gateway

In order for the client device to access external networks, the router's internal IP address was configured as the default gateway. This ensured that any traffic from the client machine intended for external networks would be routed through the Ubuntu router.

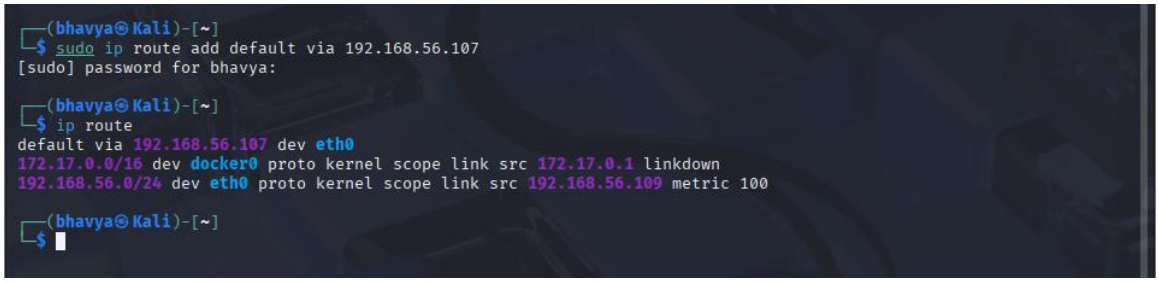


Figure 4: Routing table showing the router configured as the default gateway.

Testing Network Connectivity

After configuring the network, connectivity between the client and the router was tested using the ping command. Successful responses confirmed that the devices were able to communicate within the internal network.

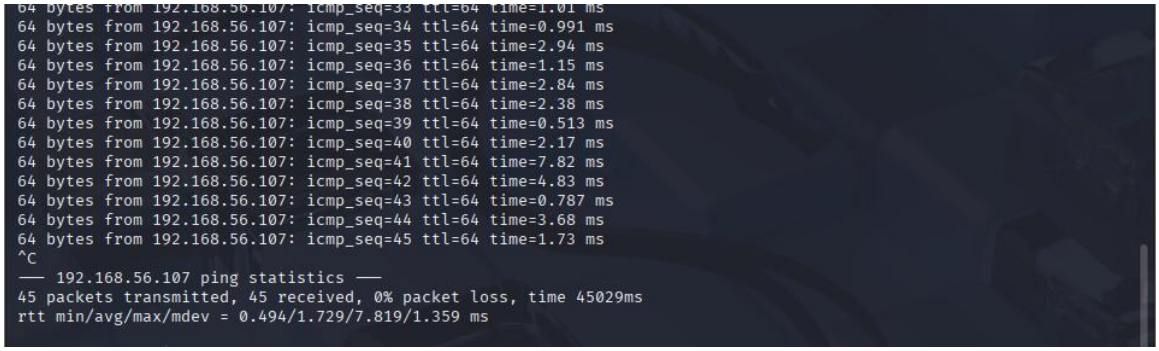


Figure 5: Ping test confirming communication between the client and the router.

Internet connectivity was also tested by sending packets to a public DNS server.

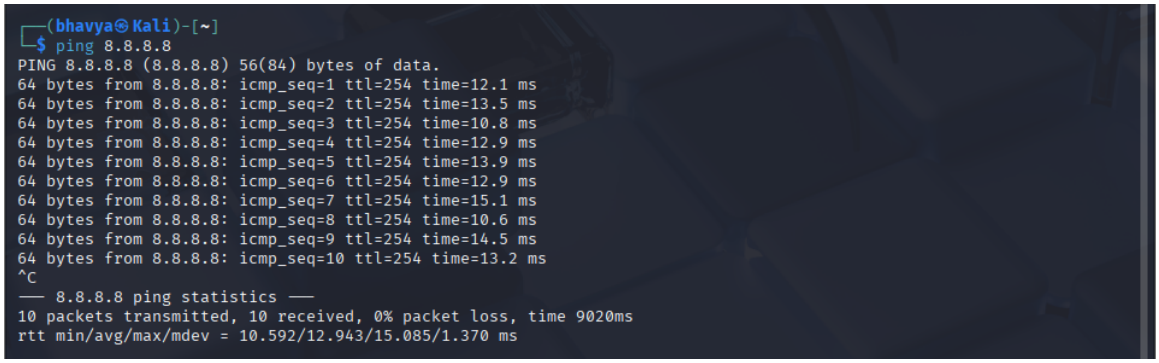


Figure 6: Successful ping responses confirming internet connectivity from the client machine.

Implementing Basic Security Measures

A firewall was enabled on the router to protect the network from unauthorized access. The firewall used was UFW, which allows administrators to control network traffic through configurable rules. Enabling the firewall helps prevent unauthorized connections from external sources.

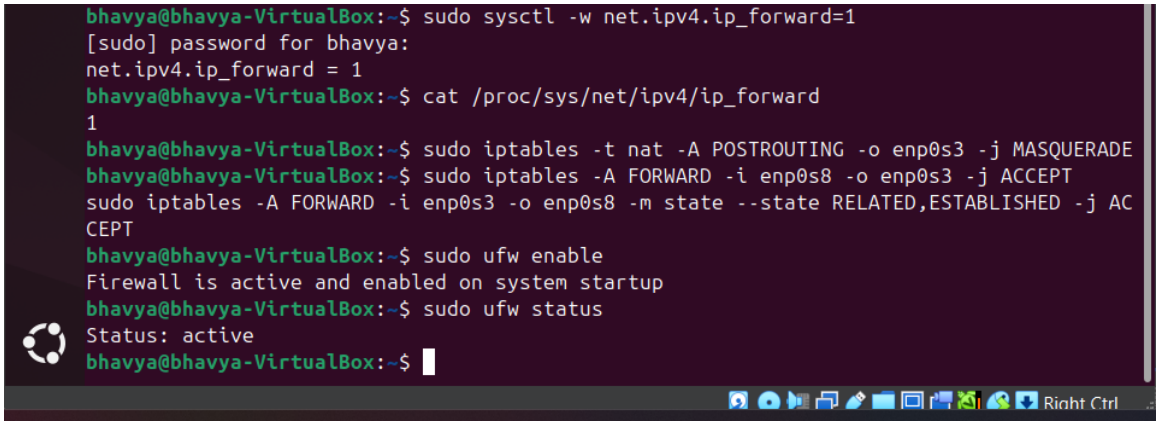


Figure 7: Firewall enabled on the router to block unauthorized network access.

In addition to enabling the firewall, several basic security configurations were considered. Default passwords for systems should be replaced with strong passwords to prevent unauthorized access. Network encryption such as WPA2 or WPA3 should also be enabled on wireless networks to protect data transmitted between devices. These practices help strengthen the overall security of the network.

Monitoring Network Traffic

Network traffic was monitored using Wireshark, which is a tool used to capture and analyze packets traveling across a network. Wireshark helps identify different types of network communication and assists in detecting potential security threats.

First, Wireshark was opened and the active network interface was selected to start packet capture.

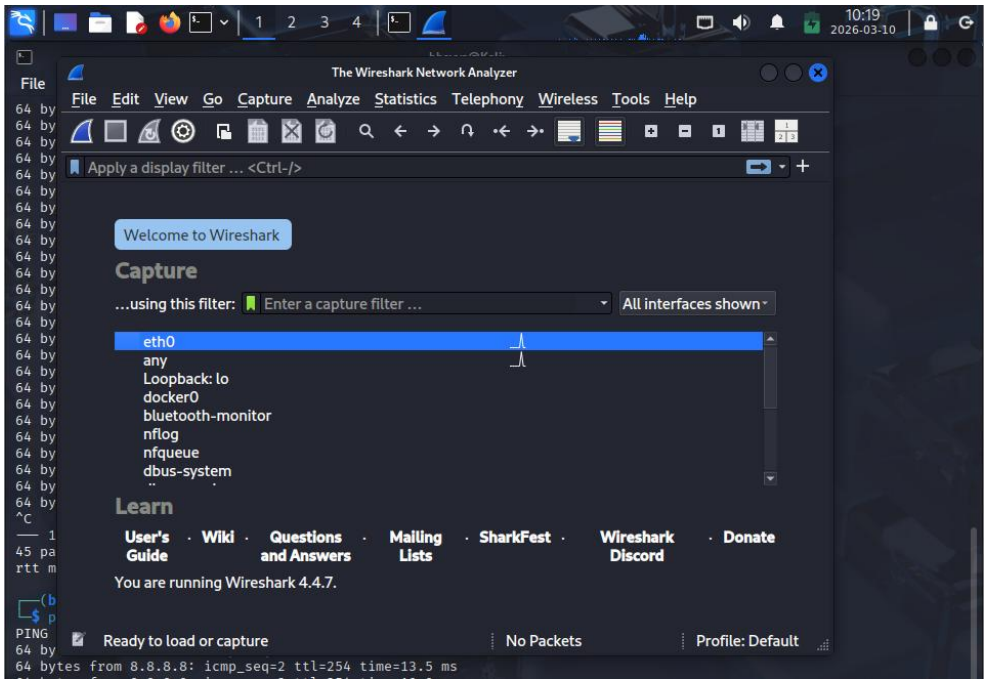


Figure 8: Wireshark interface selection showing available network interfaces.

DNS Traffic Analysis

DNS packets were captured while resolving domain names. DNS traffic shows how the system translates domain names such as website addresses into IP addresses that computers can understand.

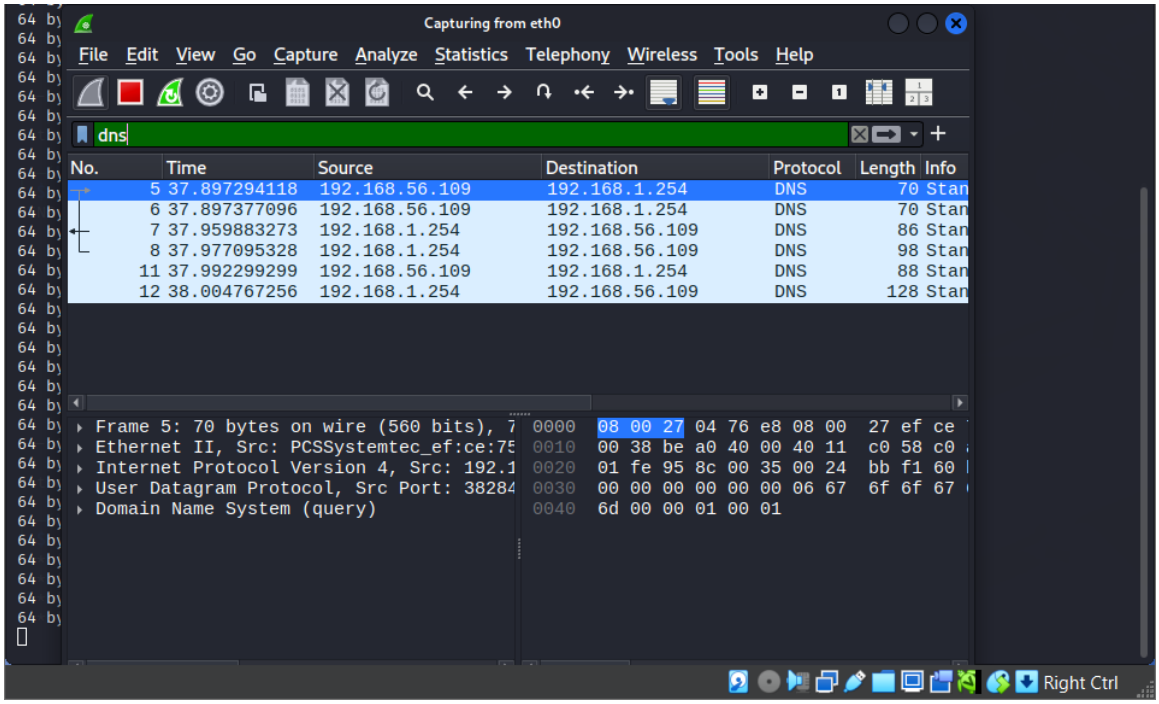


Figure 9: DNS packets captured while resolving domain names.

ICMP Traffic Analysis

ICMP packets were observed when performing ping tests. These packets are used for network diagnostics and help determine whether devices are reachable on a network.

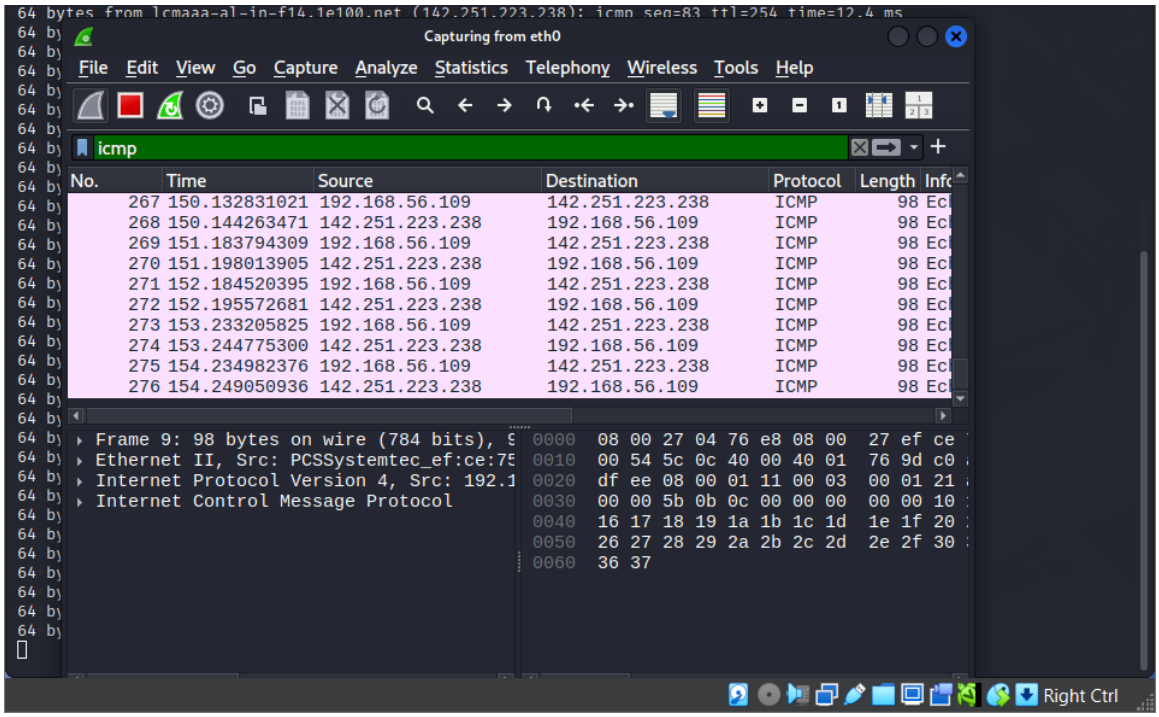


Figure 10: ICMP packets captured during the ping test.

TCP Traffic Analysis

TCP packets were observed during web communication activities. TCP is one of the main protocols used for reliable data transmission across networks, particularly for web browsing and other internet services.

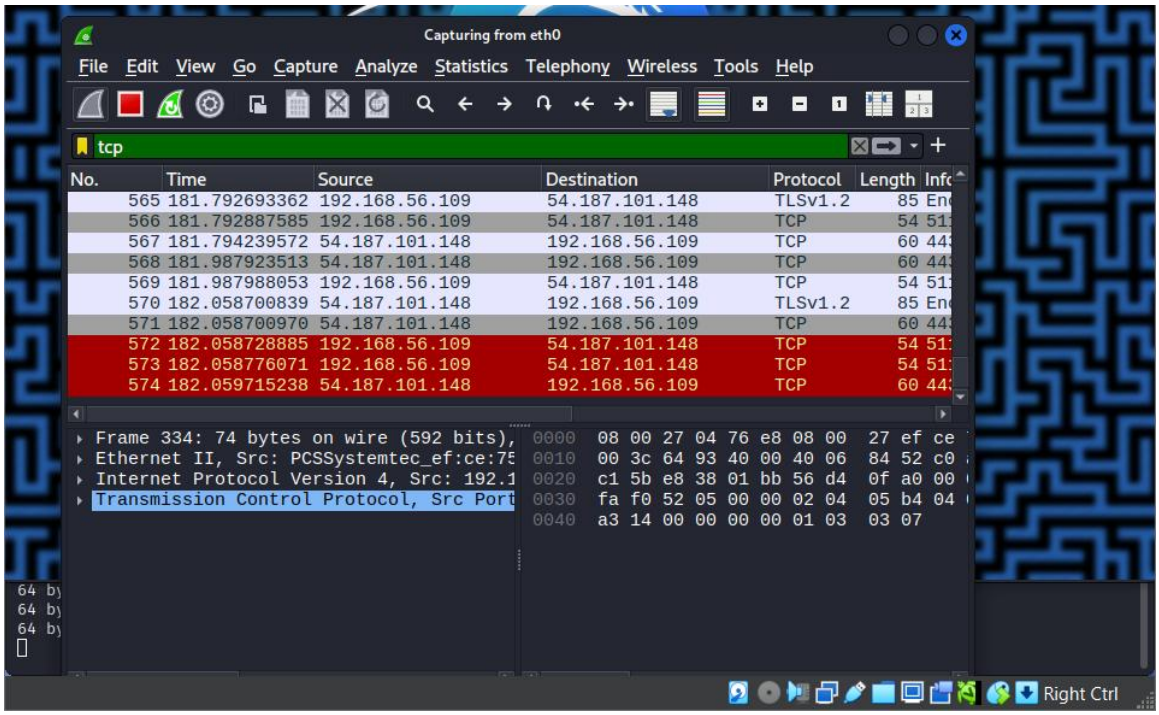


Figure 11: TCP packets captured during network communication.

Discussion and Findings

During this task, multiple types of network traffic were observed including DNS, ICMP, and TCP packets. DNS packets were used to resolve domain names into IP addresses, ICMP packets were used to test connectivity, and TCP packets were responsible for reliable data transmission during internet communication.

The firewall enabled on the router helped protect the network by filtering incoming traffic and preventing unauthorized access. Monitoring network traffic using Wireshark provided insights into how devices communicate across a network and how suspicious or abnormal traffic could potentially be identified.

Overall, implementing basic security measures such as enabling a firewall, using secure passwords, and monitoring network traffic plays an important role in protecting networks from common cybersecurity threats.

Challenges Faced and How I Overcame Them

Understanding Packet Details:
At first, the packet data in Wireshark looked very complicated. I overcame this by learning how to use filters and focusing only on important fields like source IP, destination IP, and protocol type.

Identifying Suspicious Traffic:
It was difficult to understand what counts as unusual traffic. I researched common signs of suspicious behavior, such as repeated failed connections or unexpected outbound communication.

Firewall Configuration Confusion:
I was initially confused about inbound and outbound rules. I solved this by reading documentation and testing settings to see how they affected connectivity.

Results and Outcomes

Through this task, I gained a clear understanding of basic network security concepts and common threats. I successfully configured a firewall and secured my router settings. I learned how to capture and analyze live network traffic using Wireshark and identify protocols such as DNS, HTTP, HTTPS, and TCP.

Overall, this task helped me build a strong foundation in network security and improved my confidence in handling basic security configurations and traffic analysis.

Reflection on Security Best Practices

Additional Security Measures for Larger Networks

In larger and more complex networks, stronger security controls are required to protect systems and data. Some important measures include:

- **Network segmentation** to divide the network into smaller sections and limit the spread of attacks.
- **Intrusion Detection and Prevention Systems (IDS/IPS)** to detect and respond to suspicious activities.
- **Multi-factor authentication (MFA)** to add an extra layer of protection for user accounts.
- **Regular system updates and patch management** to fix known vulnerabilities.
- **Advanced firewalls and continuous monitoring** to control and analyze network traffic.

Educating Others About Network Security

To promote safe online practices, people should be made aware of basic cybersecurity habits such as:

- Using **strong and unique passwords** for different accounts.
- Avoiding **suspicious emails, links, and attachments** that may contain malware.
- Keeping **devices and software updated** to reduce security risks.
- Connecting only to **secure and trusted Wi-Fi networks**.
- Understanding common threats such as **phishing and malware attacks**.

Educating users about these simple practices can help reduce the chances of cyber attacks in everyday internet usage.

2.2 Task 2: Introduction to Web Application Security

Objective of the task

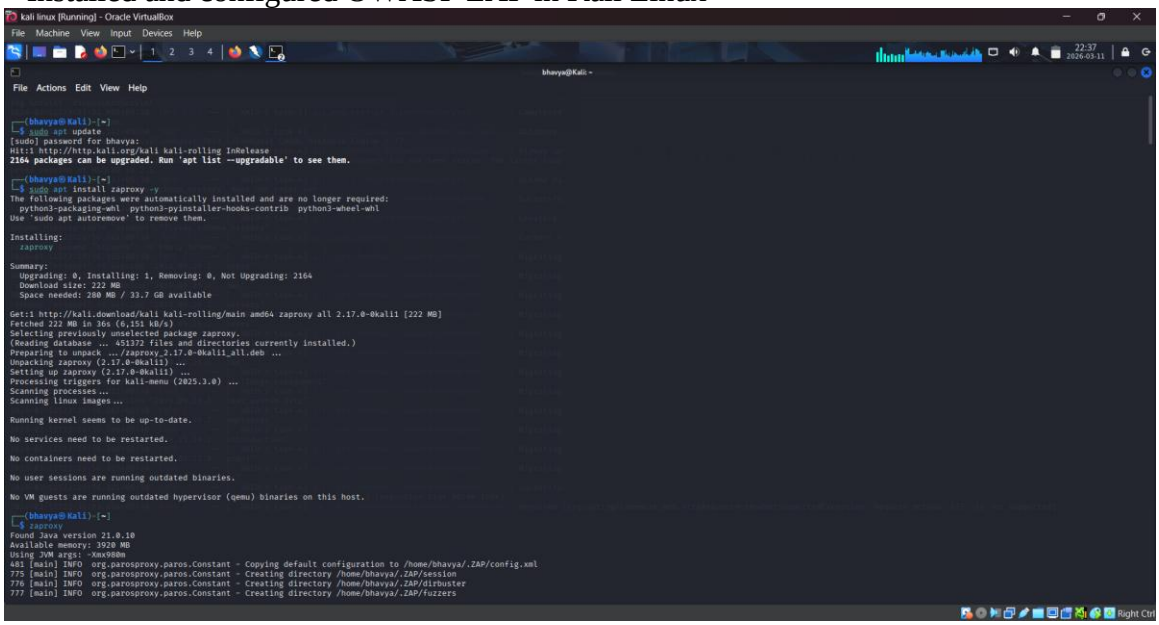
The objective of this task was to understand common web application vulnerabilities by analysing a vulnerable application. The focus was on learning how attackers exploit weaknesses such as SQL Injection, Cross-Site Scripting (XSS), and Cross-Site Request Forgery (CSRF), and gaining practical exposure to identifying and testing these vulnerabilities.

Tools and methods used

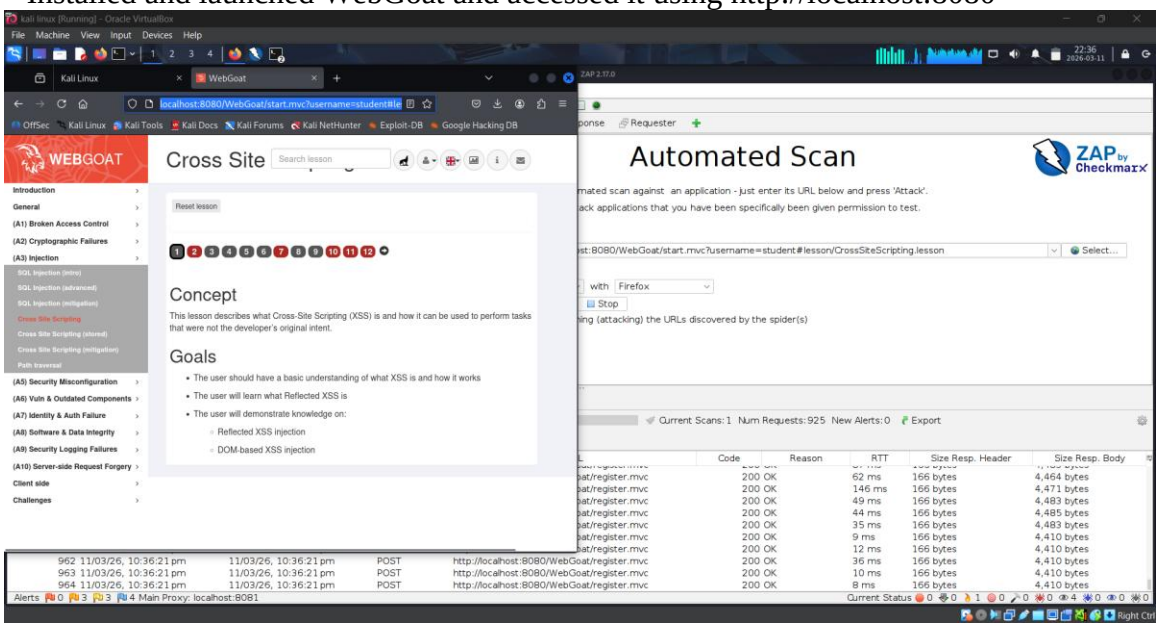
- OWASP ZAP for automated vulnerability scanning and analysis
- WebGoat as a deliberately vulnerable web application
- Kali Linux environment for setup and testing
- Manual testing techniques for exploiting vulnerabilities

Steps taken to complete the task

• Installed and configured OWASP ZAP in Kali Linux



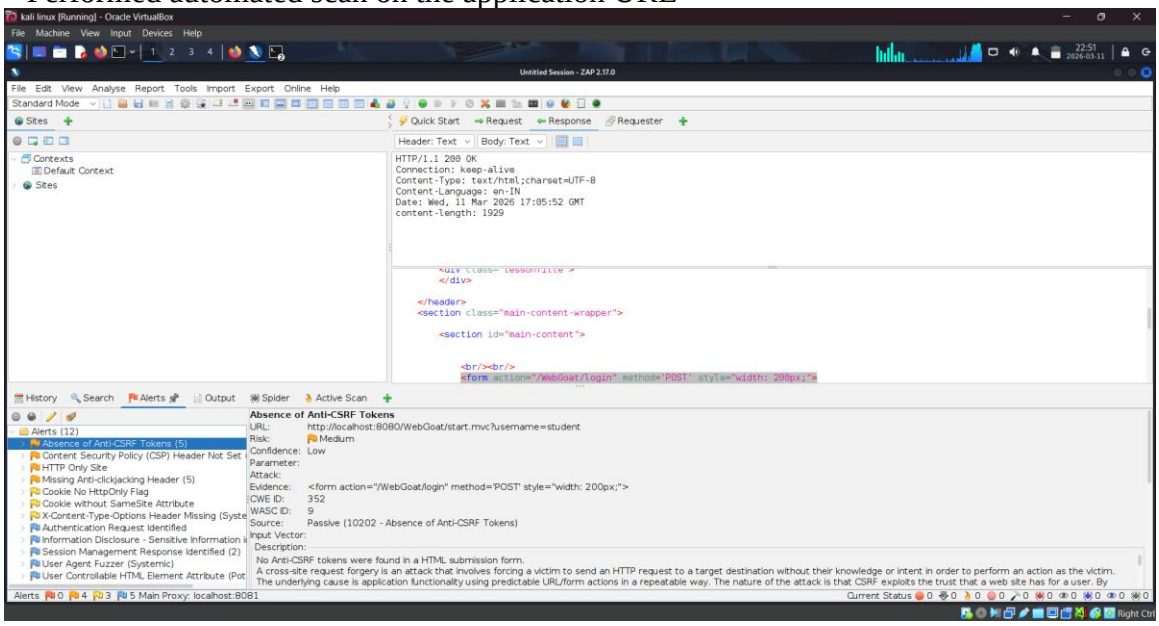
• Installed and launched WebGoat and accessed it using http://localhost:8080



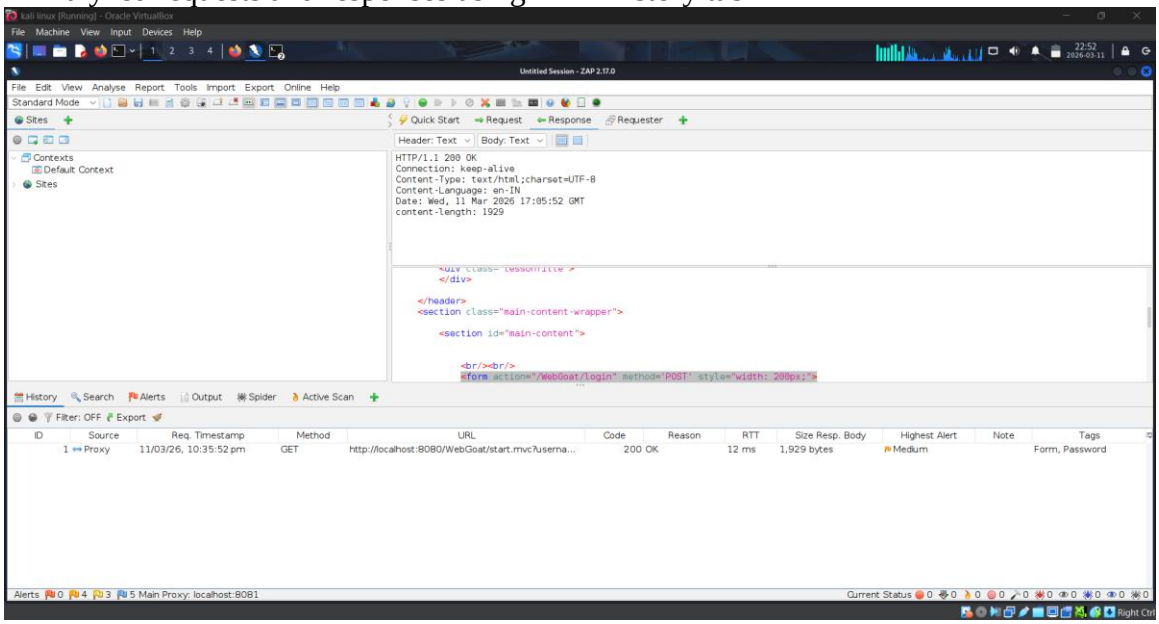
• Explored different lessons in WebGoat related to vulnerabilities

• Configured OWASP ZAP to scan the WebGoat application

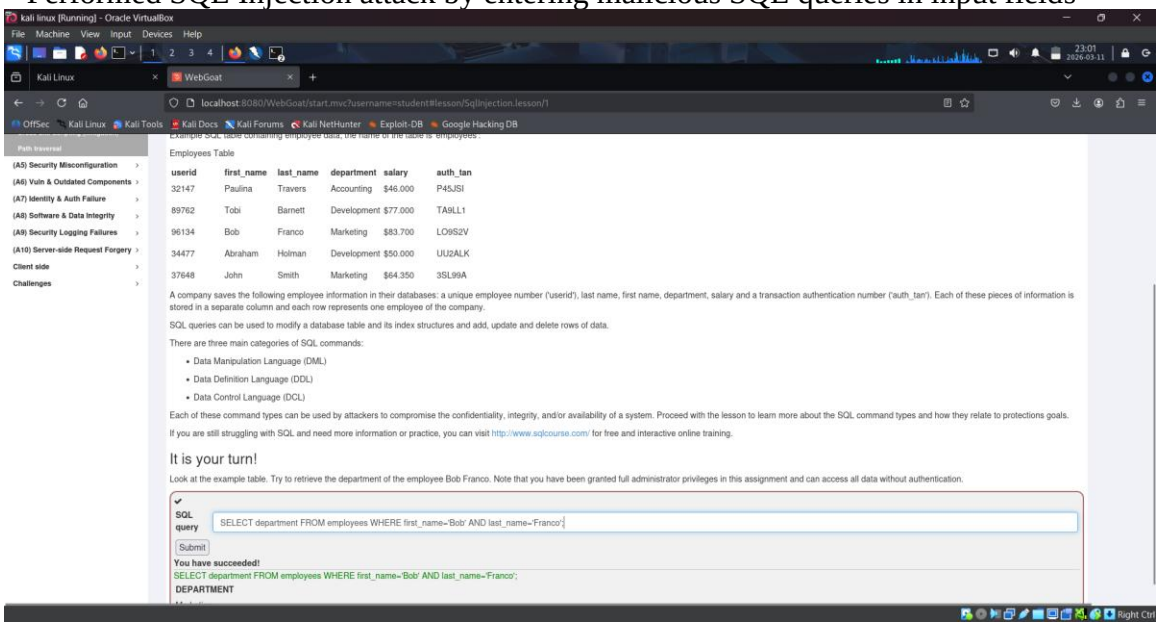
• Performed automated scan on the application URL



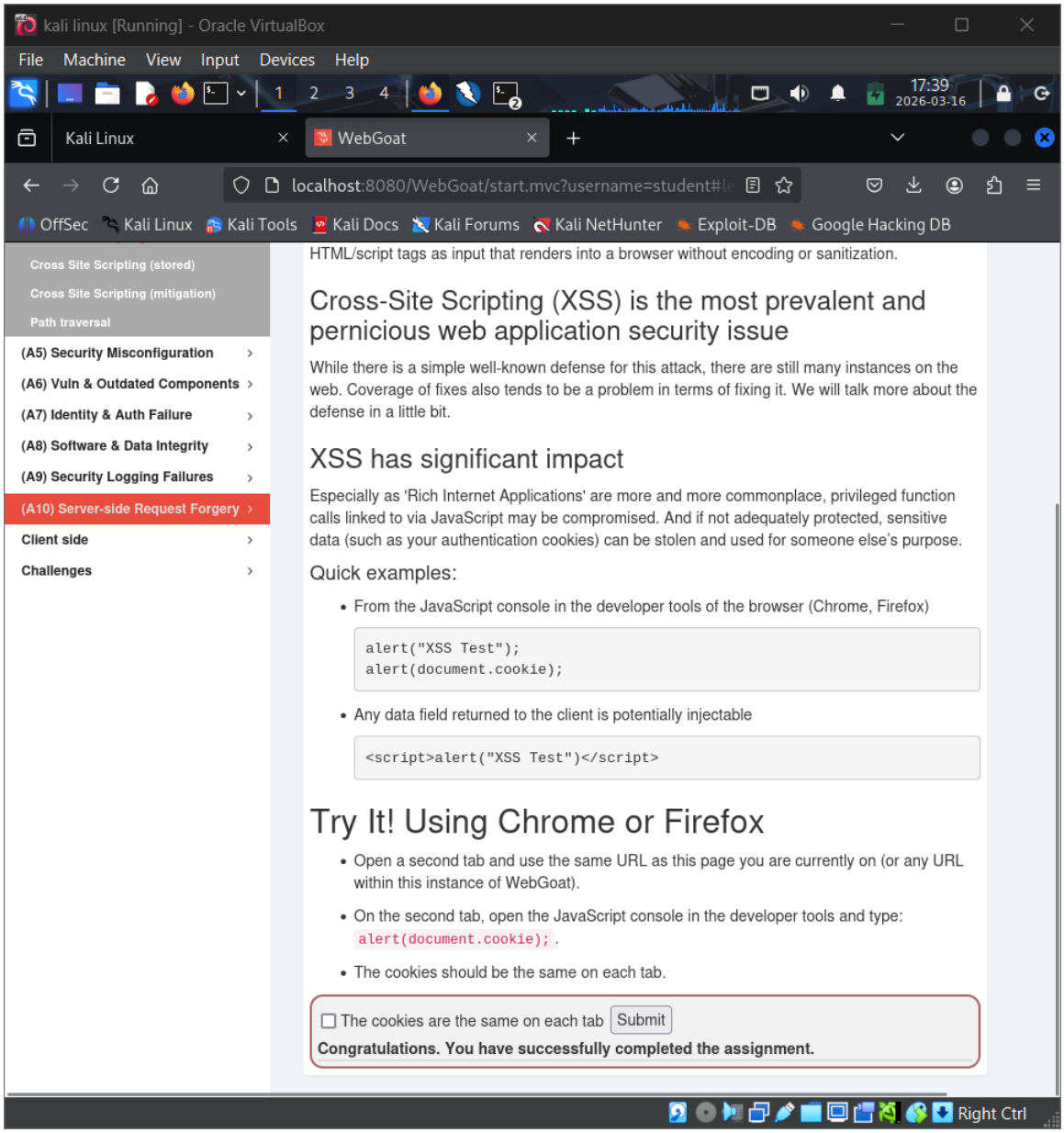
• Analyzed requests and responses using ZAP History tab



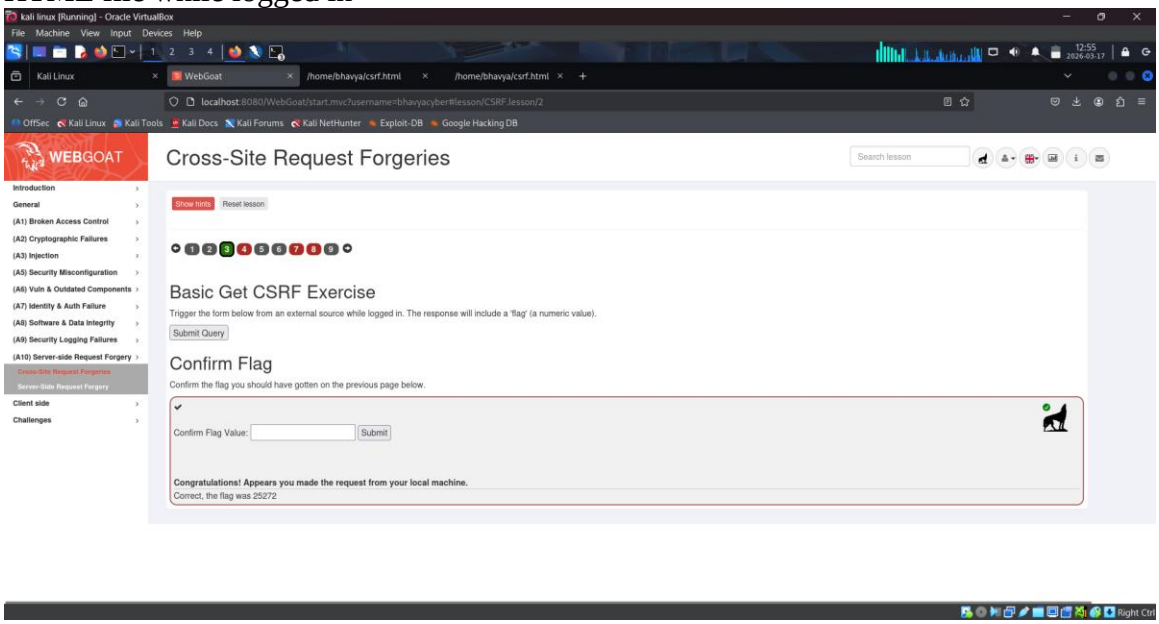
• Performed SQL Injection attack by entering malicious SQL queries in input fields



• Performed Cross-Site Scripting (XSS) by injecting JavaScript code and observing alert execution



• Performed Cross-Site Request Forgery (CSRF) by triggering a request using an external HTML file while logged in



• Documented all findings with proper screenshots and observations

Challenges faced and how they were overcome

- Faced Java compatibility issues while running WebGoat, which were resolved by installing the correct Java version
- Encountered port 8080 conflict, which was resolved by stopping the conflicting service or retrying with proper configuration
- Faced errors while downloading correct WebGoat version (404 error), resolved by using the correct release version
- Initial difficulty in configuring OWASP ZAP scanning, resolved through repeated attempts and understanding tool workflow
- Errors during SQL Injection attempts were fixed by correcting query syntax
- CSRF task initially showed a blank page, which was resolved by correctly creating and executing the HTML file

Results and outcomes

- Successfully identified and exploited SQL Injection, XSS, and CSRF vulnerabilities
- Understood how attackers manipulate input fields and user sessions
- Learned how OWASP ZAP helps in identifying vulnerabilities through automated scanning
- Gained practical experience in web application security testing
- Developed problem-solving skills while troubleshooting setup and execution issues

3. Conclusion

This task provided hands-on experience in identifying and exploiting common web application vulnerabilities. It helped in understanding how real-world attacks such as SQL Injection, XSS, and CSRF work. The use of tools like OWASP ZAP improved my ability to analyze web applications for security flaws. Overall, this task enhanced my practical knowledge and strengthened my foundation in cybersecurity concepts

4. Closing Remarks

This task was highly useful in bridging the gap between theoretical knowledge and practical implementation. It provided a clear understanding of web vulnerabilities and their impact. For improvement, more guided troubleshooting steps could be included for beginners. Overall, the experience was very informative and valuable for building a career in cybersecurity.